

IAC-26,D2,2,12,x114322

Development of an Actively Stabilized Marine Launch Platform for Small Rocket Launch Operations

Álvaro Pérez Mora^a, Aarón David Salas Alvarado^a, William Smith Hernández^a

^a School of Electronics Engineering, Technical Institute of Costa Rica (ITCR), Cartago, Costa Rica.

* Corresponding author

Abstract

Costa Rica is a small country of around 51,100 square kilometers of area, and it has extensive protected regions and scarce flat terrain, which constrains ground-based launch infrastructure and motivates alternative launch architectures. The MARLUP (Marine Launchpad) project proposes an actively stabilized floating launch platform for small rockets (motor classes A through H). The launch platform is based on a Stewart-type mechanism, with three linear actuators that control the roll, pitch and vertical displacement (heave) of the top plate, and mitigate the dominant disturbances induced by waves and tide. The dynamics were formulated from rigid-body mechanics as a nonlinear multi-input multi-output system and cast in state-space form. A digital twin was implemented in MATLAB/Simulink with Simscape Multibody from CAD geometry, including noisy, quantized and sampled sensor models and calm-sea disturbances generated from a Pierson-Moskowitz spectrum with directional spreading. Since the simplified analytic model used in earlier stages misrepresented the effective actuator geometry and inertia of the closed-chain mechanism, the force-to-acceleration map of the platform was identified directly from the digital twin through four short simulations and validated against an independent superposition prediction. On the identified model, an integral linear quadratic regulator with closed-form weights places the closed-loop bandwidth five to six times above the wave spectral peak, and a steady-state Kalman filter reconstructs the unmeasured states from the three sampled and quantized measurements. The controller and the estimator were discretized, generated as embedded C code and executed in a Processor-in-the-Loop configuration on an embedded microcontroller, while the digital twin was simulated on a host computer, and the resulting trajectories were verified against the equivalent Software-in-the-Loop execution. Compared with the previous decentralized and model-based control stages, the proposed strategy raises the wave attenuation at the spectral peak from 6 dB to between 31 and 38 dB, settles within approximately 10 s with minimal overshoot while holding the commanded heave, and reduces the simulation cost from hours to minutes. These results demonstrate the feasibility of an actively stabilized floating launch platform and establish a validated digital twin and control architecture for future experimental work and maritime launch infrastructure in emerging space nations.

Nomenclature

α, θ	roll and pitch angles of the plate (rad)
z	heave, plate height (m)
$\mathbf{q}$	generalized coordinates $[\alpha \ \theta \ z]^\top$
$\mathbf{x}, \hat{\mathbf{x}}$	state vector $[\alpha \ \dot{\alpha} \ \theta \ \dot{\theta} \ z \ \dot{z}]^\top$ and its estimate
$\mathbf{F}$	actuator force vector $[F_1 \ F_2 \ F_3]^\top$ (N)
$\mathbf{F}_{\text{trim}}$	static equilibrium force vector (N)
$\mathbf{W}$	force-to-acceleration map (3×3)
$\mathbf{c}$	gravitational bias acceleration
$\mathbf{M}(\mathbf{q}), \mathbf{J}(\mathbf{q})$	generalized inertia matrix, inverse-kinematics Jacobian
ξ	integral state
H_s, T_p, ω_p	significant wave height, peak period,

$\omega_n, \zeta, \omega_i$	peak frequency closed-loop natural frequency, damping ratio, integrator pole
u_z	vertical direction cosine of an actuator
T_s	controller sampling period (s)
$\mathbf{A}_d, \mathbf{B}_d$	zero-order-hold discretization of $\mathbf{A}, \mathbf{B}$
$\mathbf{L}_\infty$	steady-state Kalman gain
$\mathbf{Q}_k, \mathbf{R}_k$	process and measurement noise covariances

Acronyms/Abbreviations

Additive white Gaussian noise (AWGN), attitude and heading reference system (AHRS), degree of freedom (DOF), Hardware-in-the-Loop (HiL), Kalman filter (KF),

linear quadratic regulator (LQR), Marine Launchpad (MARLUP), multi-input multi-output (MIMO), Pierson–Moskowitz (P–M), Processor-in-the-Loop (PiL), root mean square (RMS), Software-in-the-Loop (SiL), zero-order hold (ZOH).

1. Introduction

Access to launch infrastructure remains one of the main barriers for emerging space nations. Ground-based launch sites require large uninhabited areas, safety exclusion zones along the flight corridor and stable, flat terrain. Costa Rica combines a small national territory, a large fraction of land under environmental protection and a mountainous topography, which makes a conventional launch range, even for small suborbital vehicles, difficult to justify. Launching from the sea removes most of these constraints, since the launch site can be relocated to open water where range safety is easier to guarantee.

Sea-based launch has precedents at very different scales, from the fixed San Marco platform off the coast of Kenya to the floating Sea Launch system and the more recent ship-based launches of small orbital vehicles [?, ?]. For small rockets, however, the floating base is itself a moving support: wave-induced roll, pitch and heave perturb the alignment of the launch rail and the initial attitude of the vehicle at lift-off. Parallel mechanisms derived from the Stewart platform [?, ?, ?] are the established solution for compensating such motions, and active motion compensation is routinely used offshore, for example in motion-compensated gangways for wind turbine access [?]. A reduced three-actuator architecture that controls only roll, pitch and heave addresses the motions that dominate launch alignment while keeping mass, cost and power consumption compatible with a student-led project.

The MARLUP project, developed within TECSpace at the Instituto Tecnológico de Costa Rica, follows this approach. The hexagonal floating base was sized from buoyancy and stability criteria [?]. A first Simscape Multibody model of the stabilization mechanism was controlled with one decentralized PID loop per axis, under sinusoidal waves and ideal sensors, and required several hours of computation for a few seconds of response [?]. A second stage replaced the PID loops with a linear quadratic regulator (LQR) and a KF designed on a simplified analytic model, added a P–M wave model and non-ideal sensors, and executed the controller on a microcontroller in a PiL configuration; applied to the multibody digital twin, however, this controller achieved only marginal stability for about 20 s [?]. The recommendation of that stage was to identify the simulated plant instead of assuming it.

This paper reports the resulting design. Its contributions are a digital twin of the platform with realistic disturbances and sensing, a diagnosis of why the rigid-body simplification commonly used for such platforms fails for a closed-chain mechanism whose legs are comparable in size to the plate, a non-invasive identification procedure that is exact at the operating point and is validated by an independent prediction, an integral LQR with closed-form weights whose bandwidth is placed deliberately with respect to the wave spectrum, and the execution of the discretized controller and estimator in PiL, verified against the equivalent SiL execution of the same generated code. Section 2 describes the system and its environment, Section 3 the dynamic model, Section 4 the identification, Section 5 the control and estimation design, Section 6 the PiL implementation, Section 7 the results, and Section 8 the conclusions.

2. System description

2.1 Launch platform concept

The MARLUP concept consists of a hexagonal floating base with six lateral floats and a central float, sized by Archimedes' principle to carry the structure, the launch plate and the lift-off load with a 25% margin [?]. The stabilization mechanism, shown in Fig. 1, is mounted on the floating base and carries the launch plate, which must remain level and at a commanded height during launch preparation and lift-off. The operating envelope is restricted to calm-sea launch windows. **[TODO: state the design sea state (H_s range) and the target rocket envelope (mass, rail length).]**

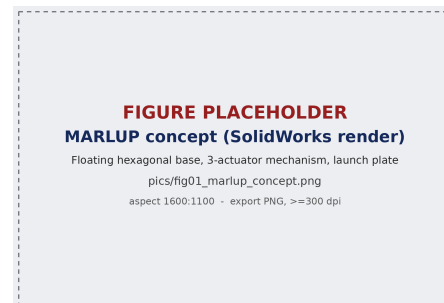


Fig. 1: Stabilization mechanism of MARLUP: launch plate, three linear actuators and the triangular support that attaches the mechanism to the floating base (SolidWorks model). **[TODO: replace by, or complement with, a render of the complete concept including the hexagonal floating base.]**

2.2 Stabilization mechanism

The moving platform is a circular plate of mass $M = 67.7$ kg carried by three linear actuators (Fig. 2). The lower anchors lie on a small inverted triangular pyramid ($H = 0.11$ m, circumscribed radius $R_b = 0.13$ m) attached to the floating structure through a four-DOF joint. The upper anchors are distributed on a circle of radius $R_t = 0.65$ m at azimuths $\varphi_i = \{-30^\circ, 90^\circ, 210^\circ\}$, and the plate rests at $z_0 = 1.5716$ m in the assembled configuration. The actuators are modeled after a double-acting hydraulic cylinder (Bosch Rexroth CDH1 MP3, 250 bar, 12.5 cm^2 piston area, ± 0.28 m usable stroke) and are represented in the digital twin as ideal force sources. The controlled coordinates are $\mathbf{q} = [\alpha \ \theta \ z]^\top$.

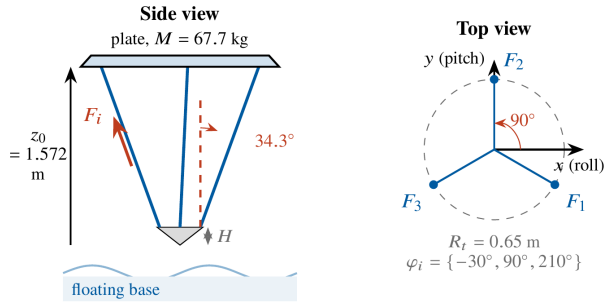


Fig. 2: Geometry of the stabilization mechanism (schematic, not to scale): side view with the measured actuator inclination of 34.3° from vertical, and top view of the plate anchors.

2.3 Sensing

The measured outputs are the two tilt angles and the plate height; no velocity is measured, which is what makes the state estimator of Section 5.4 necessary. Roll and pitch are measured by an AHRS model based on the LORD MicroStrain 3DM-GX5, and heave by an ultrasonic range sensor model based on the MaxBotix MB1040, both mounted on the plate and referred to the base. Each channel is represented in the digital twin by the same three-stage model: the ideal signal taken from the multibody sensor, additive white Gaussian noise of standard deviation σ , uniform quantization of step Δ , and zero-order-hold sampling at $T_s = 10$ ms, the period of the PiL target. The chain is deliberately kept in this order, since quantizing after adding noise reproduces the dithering that a real converter experiences and avoids the artificially periodic error that quantizing a noiseless signal would produce.

Table 1 lists the parameters. The angular standard deviation of 0.5° and the heave standard deviation of 3 mm are

the values used both in the noise blocks of the digital twin and in the measurement covariance of the estimator, so that the filter is tuned to the noise it actually receives; the consequences of violating this correspondence are discussed in Section 5.4. **[TODO: confirm the quantization steps actually set in the Simulink blocks and the seeds of the noise generators; add the datasheet values of both sensors for comparison.]**

Table 1: Sensor models used in the digital twin.

Quantity	Sensor model	σ	Δ
Roll α	AHRS (3DM-GX5)	0.5° (8.7×10^{-3} rad)	[TODO:]
Pitch θ	AHRS (3DM-GX5)	0.5° (8.7×10^{-3} rad)	[TODO:]
Heave z	Ultrasonic (MB1040)	3 mm	[TODO:]
All channels sampled with ZOH at $T_s = 10$ ms.			

2.4 Wave disturbance model

The sea state is generated from the P-M spectrum [?,?],

$$S_\eta(\omega) = \frac{5}{16} \frac{H_s^2 \omega_p^4}{\omega^5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right], \quad (1)$$

combined with a $\cos^2$ directional spreading function. The peak period $T_p = 10$ s gives $\omega_p = 2\pi/T_p = 0.628$ rad/s, the frequency that governs the control specification of Section 5. The free surface is reconstructed as a finite sum of components with random phases (Fig. 3); its elevation at the origin defines the heave of the base, and its spatial gradients define the base roll and pitch, which are imposed through the four-DOF joint. The same realization, obtained by fixing the phase seed, is reused in every comparison so that the SiL and PiL runs of Section 6 see an identical disturbance. **[TODO: state H_s , number of wave components and seed.]**

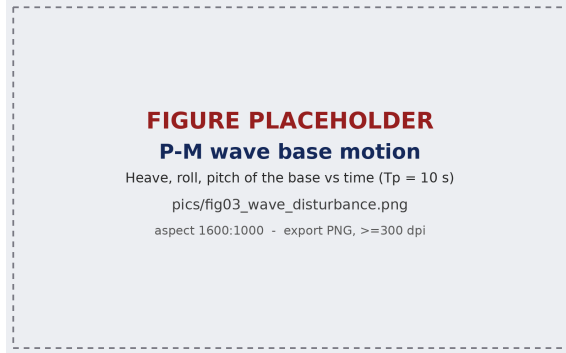


Fig. 3: Free-surface snapshot generated by the P-M model with directional spreading ($T_p = 10$ s). The disc marks the platform footprint, whose elevation and slope define the heave, roll and pitch imposed on the floating structure. **[TODO: consider adding a second panel with the time histories of base heave, roll and pitch.]**

3. Dynamic model and digital twin

3.1 Rigid-body formulation

For a rigid multibody mechanism with independent generalized coordinates $\mathbf{q}$, Lagrangian mechanics gives [?]

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{J}(\mathbf{q})^T \mathbf{F}. \quad (2)$$

The generalized inertia matrix $\mathbf{M}(\mathbf{q}) = \sum_k [\mathbf{J}_{v,k}^T m_k \mathbf{J}_{v,k} + \mathbf{J}_{\omega,k}^T \mathbf{I}_k \mathbf{J}_{\omega,k}]$ contains every moving body (plate, upper triangle and leg segments). The vector $\mathbf{h}$ collects gravity and the Coriolis and centrifugal terms, the latter quadratic in $\dot{\mathbf{q}}$. The inverse-kinematics Jacobian $\mathbf{J}(\mathbf{q}) = \partial \ell / \partial \mathbf{q}$, with ℓ the leg lengths, has rows $\mathbf{u}_i^T \partial \mathbf{p}_i / \partial \mathbf{q}$, where $\mathbf{u}_i = (\mathbf{p}_i - \mathbf{b}_i) / \ell_i$ is the unit vector from the base anchor $\mathbf{b}_i$ to the plate anchor $\mathbf{p}_i$. Equation (2) defines a nonlinear, configuration-dependent MIMO plant.

Evaluated at the assembled pose $\mathbf{q}_0$ with the plate at rest, $\mathbf{M}$, $\mathbf{h}$ and $\mathbf{J}$ become constant and every velocity-dependent term vanishes identically, so that

$$\ddot{\mathbf{q}} = \mathbf{W} \mathbf{F} + \mathbf{c}, \quad \mathbf{W} = \mathbf{M}(\mathbf{q}_0)^{-1} \mathbf{J}(\mathbf{q}_0)^T, \quad \mathbf{c} = -\mathbf{M}(\mathbf{q}_0)^{-1} \mathbf{h}(\mathbf{q}_0, \mathbf{0}). \quad (3)$$

This affine relation between forces and accelerations is exact at that point, not a linearization; it only assumes rigid bodies and evaluation at rest, both of which hold by construction at the start of a simulation from the assembled configuration. The Grübler-Kutzbach criterion applied to the Simscape topology ($N = 14$ links, $j = 15$ joints, $\sum f_i = 15$ joint freedoms) gives $6(N - 1 - j) + \sum f_i = 3$ DOF for the plate. The yaw freedom of the base joint is unactuated and unmeasured, and lies outside the model.

3.2 Limitations of the simplified analytic model

The previous stage [?] took the plate as a free rigid body, $J_x \ddot{\alpha} = \tau_x$, $J_y \ddot{\theta} = \tau_y$, $M \ddot{z} = F_z$, with the bare-plate inertias $J_x = J_y = 6.194 \text{ kg m}^2$ and $M = 67.71 \text{ kg}$, and mapped the actuator forces to generalized efforts through a constant allocation matrix,

$$\begin{bmatrix} \tau_x \\ \tau_y \\ F_z \end{bmatrix} = \mathbf{T} \mathbf{F}, \quad \mathbf{T}_{(:,i)} = u_z \begin{bmatrix} R_t \sin \varphi_i \\ -R_t \cos \varphi_i \\ 1 \end{bmatrix}, \quad (4)$$

so that $\mathbf{W}_{\text{assumed}} = \text{diag}(J_x, J_y, M)^{-1} \mathbf{T}$. Both factors are incorrect, and independently so.

The vertical direction cosine was taken as that of the normal to a face of the base pyramid, $u_z = R_b / \sqrt{R_b^2 + 4H^2} = 0.509$, which describes a real part of the mechanism but not the direction along which an actuator pushes. The actuators act along the line joining a base anchor to a plate anchor 1.57 m above it, for which the assembled model gives $u_z = 0.826$, i.e. 34.3° instead of 59.4° from vertical, a factor of 1.62 (Fig. 4). The inertia factor is wrong because the plate is the moving platform of a closed-chain mechanism: tilting it requires all legs to change length and swing, and translates the plate laterally, so every leg body contributes to $\mathbf{M}$. With the analytic geometry, the tilt inertia consistent with the measurements of Section 4 is about 57.8 kg m^2 , nine times the bare-plate value and plausibly bracketed by the bare disc (6.2 kg m^2) and a 67.7 kg point mass at the plate height ($mL^2 = 167 \text{ kg m}^2$).

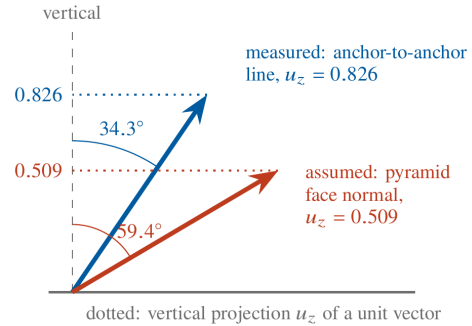


Fig. 4: Candidate actuator directions. The simplified model used the normal to a face of the base pyramid; the actuators run between anchors. Their vertical projections differ by a factor of 1.62.

Scaling the identified map back through the nominal inertias, $\mathbf{T}_{\text{ident}} = \text{diag}(J_x, J_y, M) \mathbf{W}$, allows a direct com-

parison,

$$\begin{aligned} \mathbf{T}_{\text{assumed}} &= \begin{bmatrix} -0.165 & 0.331 & -0.165 \\ -0.286 & 0 & 0.286 \\ 0.509 & 0.509 & 0.509 \end{bmatrix}, \\ \mathbf{T}_{\text{ident}} &= \begin{bmatrix} -0.017 & 0.036 & -0.018 \\ -0.031 & 0.000 & 0.030 \\ 0.835 & 0.826 & 0.816 \end{bmatrix}, \end{aligned} \quad (5)$$

which differ by 71.4% in Frobenius norm. The comparison also shows what survives: the third row of $\mathbf{T}_{\text{ident}}$ is uniform within 2.3%, as the symmetric arrangement requires, and the (2, 2) entry vanishes in both matrices, since the 90° leg has no moment arm about the pitch axis. The azimuthal structure of the earlier model was correct and only its scaling was wrong. This is the expected failure of the free-platform simplification, which is adequate for serial or lightly loaded mechanisms but not for a closed chain whose legs are comparable in mass and length to the plate.

Moreover, input–output data determine only the product $\mathbf{W} = \mathbf{M}^{-1}\mathbf{J}^\top$: assuming the bare-plate inertia yields $\mathbf{J}^\top = \mathbf{T}_{\text{ident}}$, assuming the analytic geometry yields $\mathbf{M} = \text{diag}(57.8, 57.9, 41.7)$, and both reproduce every measurement. A full derivation through (2) would require the anchor coordinates and mass properties of about eleven bodies from CAD, and would still need validation against measurements, so $\mathbf{W}$ is measured directly.

3.3 Digital twin

The digital twin was built in Simscape Multibody [?] from the SolidWorks assembly, preserving the masses, inertias and joint topology of the mechanism, and integrates the wave model and the sensor models of Section 2. Fig. 5 shows the plant subsystem: the base motion enters through the four-DOF joint at the left, the three actuators are driven by the force inputs F_1 to F_3 , and a transform sensor at the plate provides the attitude and height that feed the sensor models. The subsystem has the three actuator forces as inputs and roll, pitch, heave and their rates as outputs, with angles reported in degrees; the conversion $\mathbf{S} = \text{diag}(\pi/180, \pi/180, \pi/180, \pi/180, 1, 1)$ is applied ahead of the controller, since an unnoticed factor of 57.3 in the feedback path would act as a gain error. The model is integrated with `ode23t` at $\text{RelTol} = 10^{-4}$ and $\text{AbsTol} = 10^{-6}$; tolerances of 10^{-2} , used in earlier versions, are looser than the controlled signals themselves, which are of order 10^{-3} rad.

4. Plant identification

4.1 Experiment design

The affine map (3) has twelve unknowns, nine in $\mathbf{W}$ and three in $\mathbf{c}$, and is therefore fixed by its value at four points. Four simulations are run from rest at $\mathbf{q}_0$ with a flat sea, the feedback path opened and the allocation gain set to identity, so that a constant block injects the actuator forces directly (Fig. 6). With $\mathbf{e}_i$ the i -th unit vector and a probe force δF ,

$$\mathbf{F} = \mathbf{0} \Rightarrow \ddot{\mathbf{q}} = \mathbf{c}, \quad \mathbf{F} = \delta F \mathbf{e}_i \Rightarrow \mathbf{W}_{(:,i)} = \frac{\ddot{\mathbf{q}}_i - \mathbf{c}}{\delta F}, \quad (6)$$

and the force that holds the plate stationary follows from $\ddot{\mathbf{q}} = \mathbf{0}$ as

$$\mathbf{F}_{\text{trim}} = -\mathbf{W}^{-1}\mathbf{c}. \quad (7)$$

A fifth run with $\mathbf{F} = \delta F[1 \ 1 \ 1]^\top$ is not used for fitting: it is predicted from the other four and then measured, which makes the affine assumption falsifiable, since an engaged joint limit or an appreciable change of configuration during the window would make the prediction miss. The probe $\delta F = 400$ N is large enough for the resulting accelerations to dominate solver noise and small enough for the plate to move only microns during the measurement. Only controller-path parameters are altered, and they are restored automatically on exit, including on error; the plant subsystem is never modified.

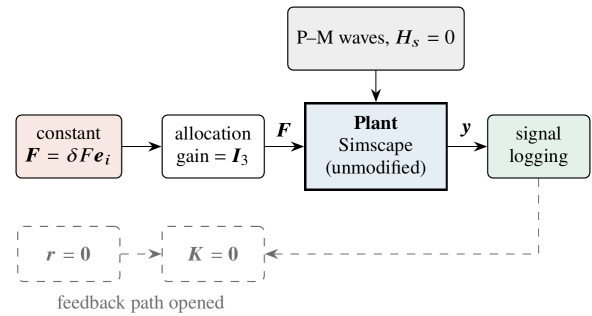


Fig. 6: Configuration for the identification runs: feedback gain and reference zeroed, allocation gain set to identity, and significant wave height set to zero so that the base is stationary.

4.2 Recovery of accelerations

The plant outputs positions and velocities. Differentiating a velocity signal numerically amplifies solver noise, and filtering it introduces a lag that biases the very quantity being measured. Under a constant force applied from rest, however, each velocity satisfies $v(t) = \dot{q}(0)t + O(t^2)$, so the acceleration is the slope at the origin. Minimizing



Fig. 5: Plant subsystem of the digital twin in Simscape Multibody: four-DOF base joint driven by the wave model, triangular support, the three actuators at $\varphi_i = \{-30^\circ, 90^\circ, 210^\circ\}$, and the transform sensor that produces roll, pitch, heave and their rates.

$\sum_k (v_k - a t_k)^2$ with the intercept constrained to zero gives

$$\ddot{q} = \frac{\sum_k t_k v_k}{\sum_k t_k^2}, \quad (8)$$

which has no tuning parameters and introduces no lag. The fit window is $t \in [4, 40]$ ms (Fig. 7): the lower bound discards four time constants of the 1 ms input filter of the physical-signal conversion, and the upper bound keeps the quadratic term negligible. These runs use ode23t with a maximum step of 0.5 ms, RelTol = 10^{-6} and AbsTol = 10^{-8} , which guarantees at least 70 samples in the window. Each run lasts 60 ms of simulated time, and the complete identification takes about 5 s of computation.

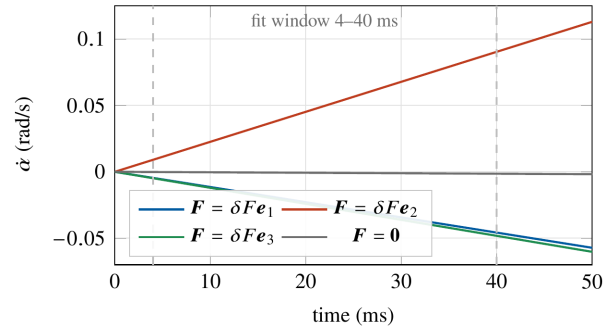


Fig. 7: Roll rate in the four identification runs. Each trace is a straight line through the origin whose slope, obtained from (8), gives one entry of $\ddot{q}$.

4.3 Identified model and validation

Table 2 lists the measured accelerations. Applying (6) and (7),

$$\mathbf{W} = 10^{-3} \begin{bmatrix} -2.772 & 5.739 & -2.924 \\ -5.027 & 0.007 & 4.872 \\ 12.336 & 12.191 & 12.054 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -0.036 \\ 0.224 \\ -9.924 \end{bmatrix}, \quad (9)$$

with rows in $\text{rad s}^{-2} \text{N}^{-1}$ (roll, pitch) and $\text{m s}^{-2} \text{N}^{-1}$ (heave), $\mathbf{F}_{\text{trim}} = [288.5 \ 273.7 \ 251.3]^\top \text{ N}$, $\det \mathbf{W} = 1.04 \times 10^{-6}$ and $\text{cond}(\mathbf{W}) = 3.02$. The free-fall bias $c_3 = -9.924 \text{ m/s}^2$ exceeds g because the unpowered plate also drags the leg linkage, a first confirmation that the linkage inertia is not negligible.

Table 2: Measured generalized accelerations (rad/s^2 , rad/s^2 , m/s^2).

$\mathbf{F}^\top$ (N)	$\ddot{\alpha}$	$\ddot{\theta}$	$\ddot{z}$	gives
[0 0 0]	-0.0360	0.2242	-9.9244	$\mathbf{c}$
[400 0 0]	-1.1448	-1.7867	-4.9902	$\mathbf{W}_{(:,1)}$
[0 400 0]	2.2596	0.2271	-5.0479	$\mathbf{W}_{(:,2)}$
[0 0 400]	-1.2055	2.1730	-5.1029	$\mathbf{W}_{(:,3)}$
[400 400 400]	-0.0187	0.1653	4.7157	check

Table 3 summarizes the validation battery. The superposition prediction, $\mathbf{W}[400 \ 400 \ 400]^\top + \mathbf{c} = [-0.0187 \ 0.1650 \ 4.7078]^\top$, matches the measured fifth run with a relative error of 1.68×10^{-3} ; since the plate then accelerates upward at 4.72 m/s^2 , the probe straddles the trim point, which is the regime in which the controller operates. The static closure compares the trim forces resolved along the identified leg direction, $(\sum_i F_{\text{trim},i}) \bar{u}_z = 813.48 \times 0.826 = 671.93 \text{ N}$, with the weight implied by the free-fall run, $M|c_3| = 671.97 \text{ N}$; because $\bar{u}_z$ is itself derived from the third row of $\mathbf{W}$, this is a consistency check bounded by the 2.3% spread of that row rather than a fully independent test. The pitch entry of the 90° leg, $W_{22} = 7.25 \times 10^{-6}$, is five orders of magnitude below the largest entry and is therefore numerically zero, as the geometry requires. Finally, the low condition number shows that $\mathbf{q}_0$ is far from a kinematic singularity. An analytic linearization of the Simscape model, in which the first Markov parameter $\mathbf{CB}$ of the velocity outputs equals $\mathbf{W}$, provides an independent algorithmic cross-check. **[TODO: report the Route C (linearize) comparison or remove this sentence.]**

Table 3: Validation of the identified model.

Check	Result	Rules out
Superposition prediction	1.68×10^{-3} rel. error	nonlinearity, active joint limits
Static equilibrium	671.93 vs 671.97 N	sign and routing errors
Zero arm of 90° leg	$W_{22}/\max W_{ij} = 5.9 \times 10^{-4}$	wrong leg assignment
Inter-leg symmetry	2.3% spread in $\mathbf{W}_{(3,:)}$	mis-assembled leg
Conditioning	$\text{cond}(\mathbf{W}) = 3.02$, rank $C = 6$	kinematic singularity

The procedure is non-invasive, reproducible and falsifiable, and requires only the `sim` command.

4.4 State-space model

With $\mathbf{x} = [\alpha \ \dot{\alpha} \ \theta \ \dot{\theta} \ z \ \dot{z}]^\top$ and the input in deviation form, $\mathbf{u} = \mathbf{F} - \mathbf{F}_{\text{trim}}$, the plant is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{y} = \mathbf{C}\mathbf{x}, \quad (10)$$

with $\mathbf{A} = \text{blkdiag}(\mathbf{A}_0, \mathbf{A}_0, \mathbf{A}_0)$, $\mathbf{A}_0 = [0 \ 1; 0 \ 0]$, $\mathbf{B}$ having zero odd rows and the rows of $\mathbf{W}$ as its even rows, and $\mathbf{C}$ selecting α , θ and z . The matrix $\mathbf{A}$ encodes only kinematics: the mechanism has no springs and negligible joint damping ($1.5 \times 10^{-4} \text{ N s/m}$), gravity enters as the constant $\mathbf{c}$ cancelled by $\mathbf{F}_{\text{trim}}$, and all cross-coupling lies in the measured off-diagonal entries of $\mathbf{W}$. The model therefore consists of three double integrators driven through a single, fully coupled input matrix. The controllability matrix has rank 6, which follows directly from $\mathbf{W}$ being nonsingular, and the pair $(\mathbf{A}, \mathbf{C})$ is observable, so the rates can be reconstructed from the three measured positions. The model is valid near $\mathbf{q}_0$, within a heave range bounded by the actuator stroke, estimated at $z_0 \pm 0.13 \text{ m}$, and a few degrees of tilt. **[TODO: verify the $\pm 0.13 \text{ m}$ range against the stroke limits.]**

5. Control and estimation design

5.1 Decoupling and allocation

Since $\mathbf{W}$ is square and well conditioned, the virtual input

$$\mathbf{v} = \mathbf{W}(\mathbf{F} - \mathbf{F}_{\text{trim}}) \implies \ddot{\mathbf{q}} = \mathbf{v} \quad (11)$$

turns the plant into three independent unit double integrators, and the actuator forces are recovered as $\mathbf{F} = \mathbf{F}_{\text{trim}} + \mathbf{W}^{-1}\mathbf{v}$. This is the role the inverse allocation $\mathbf{T}^{-1}$ played in the previous design, now with a measured rather than an assumed matrix. The condition number 3.02 bounds the amplification of identification errors by the inversion to a factor of three; for a poorly conditioned map, say $\text{cond}(\mathbf{W}) > 20$, the design should instead be carried out directly on $(\mathbf{A}, \mathbf{B})$.

5.2 Integral LQR

One integral state per controlled output, $\dot{\xi} = r - Cx$, removes the steady-state error caused by the mean component of the disturbance and by any residual mismatch in F_{trim} , which would otherwise appear directly as a standing heave offset. The augmented pair is

$$A_a = \begin{bmatrix} A & 0_{6 \times 3} \\ -C & 0_{3 \times 3} \end{bmatrix}, \quad B_a = \begin{bmatrix} B \\ 0_{3 \times 3} \end{bmatrix}, \quad (12)$$

and the controller minimizes [?]

$$\mathcal{J} = \int_0^\infty (x^\top Q x + \xi^\top Q_i \xi + u^\top R u) dt, \quad (13)$$

with Q and Q_i diagonal and $R = W^\top W$. From (11), this choice gives $u^\top R u = v^\top v$, so the controller is charged for the generalized acceleration it commands rather than for newtons. Since the effective inertias of the three axes differ by about an order of magnitude, a newton spent on heave and a newton spent on roll buy very different control authority, and weighting them equally would silently over-penalize the stiffer axis. The choice also separates the 9×9 Riccati equation exactly into three scalar-input problems. The resulting law is

$$F = F_{\text{trim}} + K_x(x_{\text{ref}} - \hat{x}) + K_i \xi, \quad (14)$$

where $\hat{x}$ is the estimate produced by the filter of Section 5.4, since only three of the six states are measured.

5.3 Bandwidth and closed-form weights

A feedback loop rejects disturbances below its bandwidth and follows them above it, so the closed-loop bandwidth must lie well above the wave peak $\omega_p = 0.628$ rad/s. The previous design sat at about 0.65 rad/s, essentially the bandwidth of the sea, and therefore tracked the waves instead of rejecting them, which alone explains its observed behavior. Each axis is now specified by $(s^2 + 2\zeta\omega_n s + \omega_n^2)(s + \omega_i) = s^3 + k_d s^2 + k_p s + k_i$, whose coefficients give the per-axis gains of the unit double integrator,

$$k_d = 2\zeta\omega_n + \omega_i, \quad k_p = \omega_n^2 + 2\zeta\omega_n\omega_i, \quad k_i = \omega_n^2\omega_i, \quad (15)$$

with the values in Table 4. The natural frequencies place the loop 4.8 and 6.4 times above ω_p ; the upper limit comes from actuator bandwidth and solver cost rather than from the algebra.

The weights are not hand-tuned: they are those for which the LQR places its poles exactly on this specification. For one decoupled axis, $\ddot{q} = v$ with $\dot{\xi} = -q$,

weights (q_α, q_d, q_i) on position, rate and integral, and unit input weight (which is what $R = W^\top W$ delivers in virtual coordinates), the transfer from v to the states is $G(s) = [1/s^2 \ 1/s \ -1/s^3]^\top$. The optimal poles are the stable roots of the return-difference equation $1 + G(-s)^\top Q_{\text{axis}} G(s) = 0$ [?], which reduces to

$$s^6 - q_d s^4 + q_\alpha s^2 - q_i = 0. \quad (16)$$

Requiring these roots to coincide with those of $\phi(s) = s^3 + k_d s^2 + k_p s + k_i$ means that (16) must equal $-\phi(s)\phi(-s) = s^6 - (k_d^2 - 2k_p)s^4 - (2k_d k_i - k_p^2)s^2 - k_i^2$, so that $q_d = k_d^2 - 2k_p$, $q_\alpha = k_p^2 - 2k_d k_i$ and $q_i = k_i^2$. Substituting (15),

$$q_\alpha = \omega_n^2 [\omega_n^2 + 2\omega_i^2(2\zeta^2 - 1)], \quad q_d = 2\omega_n^2(2\zeta^2 - 1) + \omega_i^2, \quad q_i = \omega_n^4 \omega_i^2. \quad (17)$$

Positive semidefiniteness requires $2\zeta^2 - 1 \geq 0$, i.e. $\zeta \geq 1/\sqrt{2}$, a second and independent reason for $\zeta = 0.9$. For the roll axis, $2\zeta^2 - 1 = 0.62$ gives $q_d = 11.72$, $q_\alpha = 87.28$ and $q_i = 45.56$.

Table 4: Closed-loop specification, LQR weights and equivalent per-axis gains.

Axis	ω_n	ζ	ω_i	q_α	q_d	q_i	k_p	k_d	k_i
Roll	3.0	0.9	0.75	87.28	11.72	45.56	13.05	6.15	6.75
Pitch	3.0	0.9	0.75	87.28	11.72	45.56	13.05	6.15	6.75
Heave	4.0	0.9	1.00	275.84	20.84	256.00	23.20	8.20	16.00

Solving the algebraic Riccati equation for (12) and (13) gives $K_a = R^{-1}B_a^\top P = [K_x \ -K_i]$. These gains coincide with the pole-placement construction $K_x = W^{-1}\text{blkdiag}([k_p \ k_d])$, $K_i = W^{-1}\text{diag}(k_i)$ to relative differences of 3×10^{-14} and 5×10^{-14} , and the closed-loop eigenvalues are $-2.70 \pm 1.31j$ (roll and pitch), $-3.60 \pm 1.74j$ (heave), -0.75 (twice) and -1.00 , exactly on the specification (Fig. 8). By contrast, the earlier design bisected a scalar ρ in $R = \rho I$ until the slowest pole met a bandwidth target; with integral augmentation the slow poles converge to transmission zeros fixed by the ratio Q_i/Q , which ρ does not affect, so the search ran to its lower bracket $\rho = 10^{-9}$ and returned gains of order 6×10^8 .

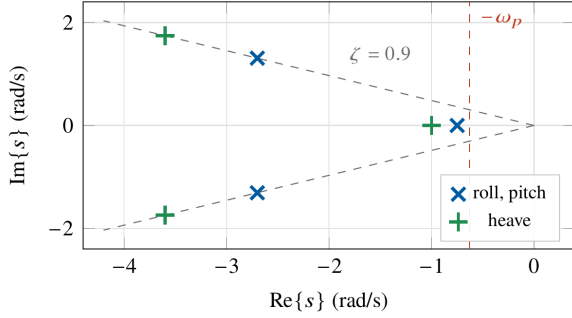


Fig. 8: Closed-loop eigenvalues. All lie on the $\zeta = 0.9$ rays or on the real axis, well to the left of the wave peak frequency.

5.4 State estimation

Only α , θ and z are measured, and they are measured with the noise, quantization and sampling described in Section 2.3, so the three rates required by (14) must be reconstructed. A discrete KF [?] is designed on the ZOH discretization of (10) at $T_s = 10$ ms. Because A is nilpotent with $A^2 = 0$, the discretization is available in closed form,

$$A_d = I + T_s A, \quad B_d = (T_s I + \frac{1}{2} T_s^2 A) B, \quad (18)$$

with no matrix exponential and no approximation, so the sampled model of each axis is the exact double integrator $[1 \ T_s; 0 \ 1]$ driven by $[T_s^2/2 \ T_s]^\top$ times the corresponding row of W . The process and measurement equations are $\mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k + \mathbf{w}_k$ and $\mathbf{y}_k = C \mathbf{x}_k + \mathbf{v}_k$, with $\mathbf{w}_k$ and $\mathbf{v}_k$ zero-mean, white and mutually uncorrelated, of covariances Q_k and R_k . The filter runs in the usual predictor–corrector form,

$$\begin{aligned} \hat{\mathbf{x}}_{k|k-1} &= A_d \hat{\mathbf{x}}_{k-1|k-1} + B_d \mathbf{u}_{k-1}, \\ \hat{\mathbf{x}}_{k|k} &= \hat{\mathbf{x}}_{k|k-1} + L_\infty (\mathbf{y}_k - C \hat{\mathbf{x}}_{k|k-1}). \end{aligned} \quad (19)$$

Three decisions define the filter. First, the measurement covariance is taken directly from the sensor models, $R_k = \text{diag}(\sigma_\alpha^2, \sigma_\theta^2, \sigma_z^2) = \text{diag}(7.62 \times 10^{-5}, 7.62 \times 10^{-5}, 9.0 \times 10^{-6})$ in rad^2 and m^2 , following Table 1. This is not a tuning knob: an overestimated R_k makes the filter distrust the measurements and integrate the model open-loop, which is the failure observed with the earlier estimator [?], while an underestimated one feeds sensor noise straight into the rate estimates and, through K_x , into the actuator commands. The quantization is not white and is not represented exactly by R_k ; adding the variance $\Delta^2/12$ of a uniform quantizer to each diagonal entry accounts for it to first order. Second, the process covariance Q_k represents what the model omits, essentially the wave-induced

forcing transmitted through the base joint and the neglected actuator dynamics. It is diagonal with the rate entries dominating the position entries, since the omitted effects act as unknown accelerations; the structurally equivalent alternative $Q_k = \Gamma \text{diag}(\sigma_{a,i}^2) \Gamma^\top$, with Γ built per axis from $[T_s^2/2 \ T_s]^\top$, encodes the same assumption with a single tuning parameter per axis. **[TODO: report the final Q_k and, if used, the $\sigma_{a,i}$.]** Third, the gain is not recomputed online: since (A_d, C) is observable and Q_k and R_k are constant, the Riccati recursion converges to a steady-state gain L_∞ , which is computed offline and implemented as a constant matrix. The estimator then costs two matrix–vector products per sample, its execution time is deterministic, and no covariance matrix has to be stored or propagated on the target, which matters directly for the PiL implementation of Section 6. **[TODO: give L_∞ and the estimator poles $\text{eig}(A_d - L_\infty C A_d)$, and check that they are three to five times faster than the controller poles.]**

Two implementation details complete the filter. The command used in the prediction step is the saturated force actually applied to the plant, not the unsaturated output of (14), since feeding the estimator a command the plant never received makes the estimate drift whenever the actuators saturate. And the measurement vector entering (19) is the one converted to radians by S , consistently with the units in which A_d , B_d and R_k are expressed. An extended filter is not required here: in deviation coordinates about $\mathbf{q}_0$ the identified model is linear by construction, and the nonlinearity of (2) only becomes relevant if the identification is repeated over the stroke, which is left as future work. **[TODO: report the estimation error of the three rates against the ideal Simscape outputs (RMS and peak) over a wave run.]**

5.5 Discrete implementation

The complete controller is discretized at $T_s = 10$ ms. The integral state is updated by forward Euler with an anti-windup correction,

$$\xi_{k+1} = \xi_k + T_s (r_k - y_k) + \beta_{aw} T_s (F_{\text{sat},k} - F_k), \quad (20)$$

with $\beta_{aw} = 1$, and the command is saturated at the cylinder force limit before reaching the plant. The design is applied to the digital twin by changing four block parameters, without structural edits: the feedback gain becomes $K_x S$, which absorbs both the state feedback and the W^{-1} allocation; the former allocation gain T^{-1} becomes the identity; the feedforward becomes F_{trim} instead of about 64 N per actuator; and the reference sets the commanded height. Fig. 9 shows the complete loop. **[TODO: confirm that the anti-windup path in (20) is the one implemented]**

and that the integrator block uses forward Euler with the same T_s .]

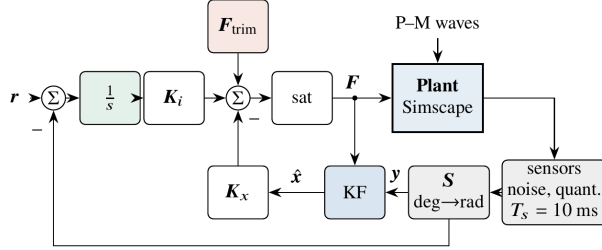


Fig. 9: Closed-loop architecture. The feedforward F_{trim} cancels gravity, K_x and K_i include the allocation W^{-1} , and the KF reconstructs the rates from the sampled measurements.

6. Processor-in-the-Loop implementation

6.1 Verification chain

The controller of Section 5 is a continuous-time design that must eventually run as compiled code on an embedded processor, and the two differ in sampling, arithmetic and execution time. These effects are separated by verifying the design in three successive configurations of the same model, which is the standard practice of model-based design [?]. In the normal simulation the controller is evaluated by the Simulink solver in host double precision. In SiL the controller subsystem is replaced by the C code generated from it, compiled for the host and executed in the same process, which verifies that the generated code reproduces the model. In PiL the same C code is cross-compiled for the target microcontroller and executed on it, while the plant, the wave model and the sensor models remain on the host; at each sampling instant the host sends the block inputs over a serial link, triggers one step of the generated code on the target and reads back the actuator commands. What changes between SiL and PiL is therefore the processor and its compiler, not the algorithm, so any discrepancy between the two is attributable to the target arithmetic and its math library rather than to the control design.

This configuration verifies functional and numerical behavior and measures resource usage on the real processor, but it does not verify real-time behavior: the host advances simulated time only when the exchange completes, so the loop runs slower than real time and no deadline is enforced. Confirming that the measured execution time fits within T_s on the target is therefore a necessary condition for real-time operation, and Hardware-in-the-Loop testing remains the step that would verify it.

6.2 Setup

Fig. 10 shows the arrangement and Fig. 11 the two Simulink configurations. The controller, the integral action and the estimator are enclosed in a single subsystem, PiL_MARLUP, which takes the reference error as input and returns the three actuator forces. That subsystem is the only element that changes between the two harnesses: it is built once from the design of Section 5 and then executed either as generated code compiled for the host, which is the SiL configuration of Fig. 11a, or as the same generated code cross-compiled for the microcontroller, which is the PiL configuration of Fig. 11b. The plant subsystem, the P-M wave generator, the sensor models and the reference are bit-for-bit identical in both, and the wave realization is fixed by its seed, so that the two runs are driven by the same disturbance sample and can be compared point by point. [TODO: target board and clock frequency, toolchain and support package, communication interface and baud rate, data type used in the generated code (double or single), and whether an FPU is available on the target.]

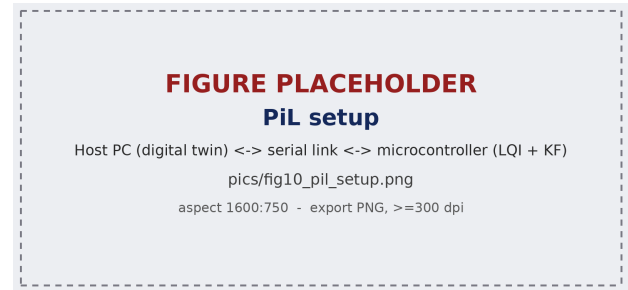


Fig. 10: PiL configuration: the host computer integrates the digital twin, the wave model and the sensors, and the microcontroller executes the generated control law and estimator, exchanging measurements and commands every T_s .

6.3 Code generation and execution metrics

The controller and estimator subsystem is configured as an atomic subsystem with a fixed sample time of T_s , and code is generated from it with the embedded real-time target, with execution-time profiling enabled so that the target reports the duration of each step. Table 5 collects the resulting metrics. The quantity that decides feasibility is the ratio of the worst-case execution time to T_s : the control law is a set of matrix-vector products of dimension at most 3×6 plus the estimator update, so its cost is dominated by about one hundred multiply-accumulate operations per step, and a comfortable margin is expected on any modern

microcontroller with hardware floating point. **[TODO: describe the code generation flow actually used (Embedded Coder target, support package, or external mode) and fill Table 5 with the profiling report.]**

Table 5: PiL execution metrics.

Metric	Value
Target microcontroller and clock	[TODO:]
Toolchain and code-generation target	[TODO:]
Sampling period T_s	10 ms
Data type of the generated code	[TODO:]
Average execution time per step	[TODO:]
Worst-case execution time per step	[TODO:]
CPU load at T_s	[TODO:]
Flash / RAM usage	[TODO:]
Max. SiL–PiL output difference	[TODO:]

6.4 SiL versus PiL

Both configurations execute the same generated code, so the comparison isolates the effect of the target processor. The residual is defined per channel as the difference between the PiL and SiL trajectories under the same wave realization, and is reported as its maximum absolute value and its RMS over the run. Two mechanisms can make it nonzero: the target may evaluate in single precision, or emulate double precision in software, where the host uses hardware double precision, and its math library may round differently. Both act at the level of machine round-off, but the integral state accumulates them, so the residual is expected to grow slowly rather than to remain bounded by a constant, and the meaningful check is that it stays several orders of magnitude below the controlled signals themselves, which are of order 10^{-3} rad and 10^{-3} m. A residual of the same order as the response, or a divergence between the two runs, would instead indicate a genuine implementation defect, such as a different sample time in the generated subsystem, an uninitialized state or a saturation applied on only one of the two paths.

Fig. 12 overlays the two responses. **[TODO: report the maximum and RMS residual per channel, state whether the growth with time is consistent with accumulation in the integral state, and comment on the execution-time margin measured in Table 5.]**

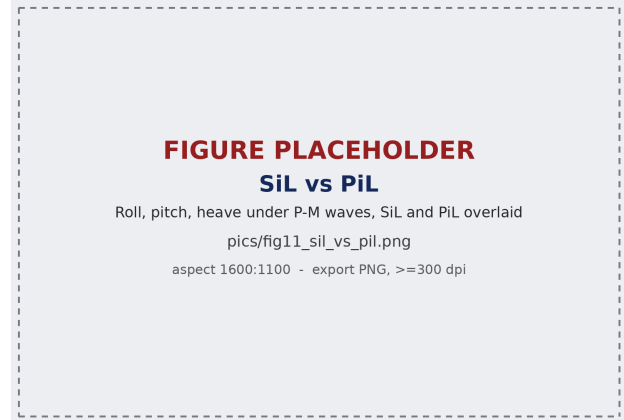


Fig. 12: Roll, pitch and heave under the same P–M wave realization with the controller executed in SiL and in PiL, and the residual between both executions.

7. Results and discussion

7.1 Disturbance rejection

For each decoupled axis the loop transfer is $L(s) = (k_d s^2 + k_p s + k_i)/s^3$, so the output sensitivity is

$$S(s) = \frac{1}{1 + L(s)} = \frac{s^3}{s^3 + k_d s^2 + k_p s + k_i}. \quad (21)$$

Its triple zero at the origin gives 60 dB/decade of attenuation at low frequency, which is the structural reason why an integral-augmented double integrator rejects slow wave motion so effectively. Fig. 13 and Table 6 compare the proposed and previous designs at the wave peak: the attenuation improves by factors of 19 and 40. This improvement is not the result of finer tuning but of placing the bandwidth deliberately, which in turn requires a correct plant model.

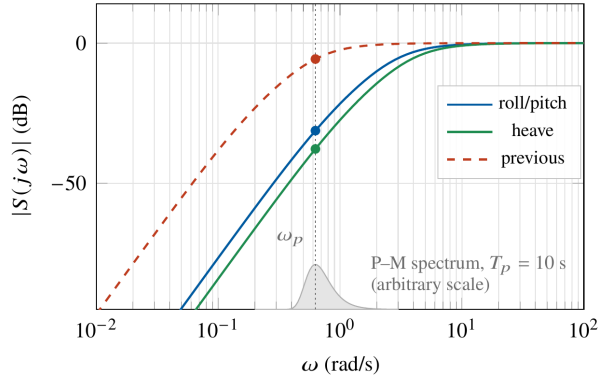


Fig. 13: Output sensitivity of the proposed design ($\omega_n = 3$ rad/s for roll and pitch, 4 rad/s for heave) and of the previous design ($\omega_n \approx 0.65$ rad/s), with the P-M spectrum shown for reference.

Table 6: Attenuation at the wave spectral peak $\omega_p = 0.628$ rad/s.

Design	$ S(j\omega_p) $	dB	Improvement
Previous, $\omega_n \approx 0.65$ rad/s	0.520	-5.7	baseline
Roll/pitch, $\omega_n = 3$ rad/s	0.027	-31.2	19×
Heave, $\omega_n = 4$ rad/s	0.013	-37.8	40×

7.2 Reference tracking and actuator authority

The loop is designed as a regulator, whose main task is disturbance rejection; reference tracking is a secondary property. With a direct reference, the numerator zero introduced by the integral action at $s = -k_i/k_p$ (-0.517 rad/s for roll and pitch, -0.690 rad/s for heave) produces a 19.9% overshoot with 2% settling times of 4.8 s and 3.6 s (Fig. 14a). A first-order prefilter that cancels this zero,

$$r_f(s) = \frac{k_i/k_p}{s + k_i/k_p} r(s), \quad (22)$$

yields a monotone response settling in 5.9 s and 4.4 s, dominated by the integrator pole. Either choice suits a constant height reference; the prefilter is preferred if the reference is slewed. For a 1° roll step the actuator forces deviate by about 13 N on legs 1 and 3 and 27 N on leg 2 from a static trim of 251–289 N (Fig. 14b), i.e. about 10% of the trim for a 1° error, the right order of magnitude for this machine, whereas the previous design demanded of order 10^8 N/rad. Over an envelope of 2° tilt, 10° /s tilt rate, 8 cm heave and 0.4 m/s heave rate, the commanded forces remain within $[-136, 669]$ N, well inside the ± 3 kN capability of the cylinder.

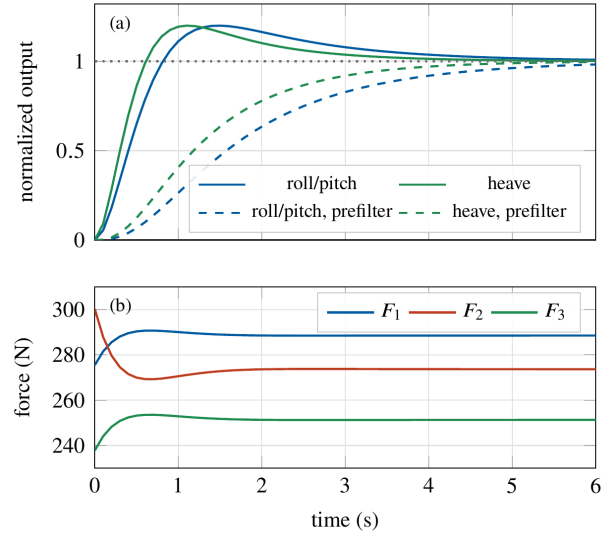


Fig. 14: (a) Closed-loop step responses with direct reference (solid) and with prefilter (dashed). (b) Actuator forces during a 1° roll reference step.

7.3 Closed-loop digital twin under waves

The design is finally applied to the full digital twin, with the P-M sea state of Section 2, the noisy and quantized sensors of Table 1 and the estimator of Section 5.4 in the loop, starting from the assembled configuration and holding a commanded height. Fig. 15 shows the response and Table 7 summarizes it. The comparison of interest is against the motion of the base itself, which is the disturbance the mechanism is meant to remove, rather than against an arbitrary reference. **[TODO: report RMS and peak roll, pitch and heave of the base and of the plate, the resulting reduction factor per axis, the settling time from the initial condition, the actuator force range and stroke usage, and the drift of the free yaw over the run.]**

Table 7: Closed-loop performance under the P-M sea state ($T_p = 10$ s).

Quantity	Base (input)	Plate (controlled)	Reduction
Roll RMS	[TODO:]	[TODO:]	[TODO:]
Pitch RMS	[TODO:]	[TODO:]	[TODO:]
Heave RMS	[TODO:]	[TODO:]	[TODO:]
Roll peak	[TODO:]	[TODO:]	[TODO:]
Pitch peak	[TODO:]	[TODO:]	[TODO:]
Heave peak	[TODO:]	[TODO:]	[TODO:]
Force range [min, max]: [TODO:] Stroke used: [TODO:]			

7.4 Comparison with previous approaches

Table 8 compares the three control stages of MARLUP. Replacing the three decentralized PID loops by a single state-feedback law reduced the simulation cost of 10 s of response from 5–13 h to about 1–2 min [?], and the identified model is what turns that computational gain into a stable and well-damped closed loop.

Table 8: Comparison of the MARLUP control stages.

	PID [?]	LQR+KF [?]	This work
Plant model	none (tuned on twin)	analytic, simplified	identified from twin
Waves	sinusoidal	P-M	P-M
Sensors	ideal	noisy, quantized	noisy, quantized
Estimator	none	KF, mistuned R_k	steady-state KF
$ S(j\omega_p) $	[TODO:]	–5.7 dB	–31 to –38 dB
Embedded execution	none	PiL	SiL and PiL
Result on twin	stable, gains up to 10^{15}	marginal, ≈ 20 s	[TODO:]
Sim. time (10 s)	5–13 h	1–2 min	[TODO:]

7.5 Limitations

The model is exact only at q_0 ; repeating the identification at other heights would convert its validity range from an estimate into a measured bound. The actuators are ideal force sources, so hydraulic dynamics, friction and bandwidth limits are not represented, which is the main reason for keeping the closed loop at 3–4 rad/s rather than faster. The double-integrator structure of A is asserted rather than identified, the Euler angles coincide with body roll and pitch only near level, and the unactuated yaw of the base joint may drift in long simulations and rotate the actuator frame away from the measurement frame. The PiL configuration verifies the arithmetic and the resource usage of the generated code but not its real-time behavior, since the host advances only when the serial exchange completes. Lift-off loads are not yet included. **[TODO: add any limitation revealed by the PiL runs.]**

8. Conclusions

An actively stabilized floating launch platform for small rockets was developed at the level of a validated digital twin and embedded controller. The analysis showed that the free-platform rigid-body simplification misrepresents a closed-chain mechanism with heavy legs by 71%, and that the map from actuator forces to accelerations can instead be identified exactly at the operating point from four

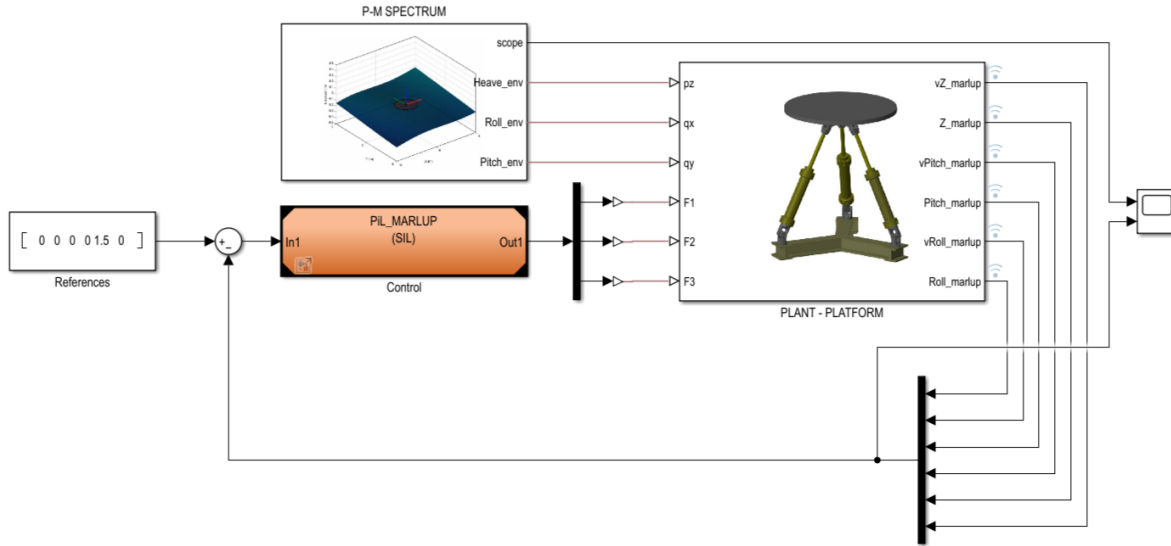
short simulations, without modifying the plant and with an independent prediction as validation. On the identified model, an integral LQR with closed-form weights places the bandwidth five to six times above the wave spectral peak and attenuates the dominant wave motion by 31 to 38 dB, with actuator demands of the correct order of magnitude, while a steady-state Kalman filter reconstructs the three unmeasured rates from noisy, quantized and sampled measurements at a fixed and deterministic computational cost. The discretized controller and estimator were generated as embedded C code and executed on a microcontroller in PiL, and verified against the equivalent SiL execution of the same code. **[TODO: summarize the PiL metrics, the SiL–PiL residual and the digital-twin wave results in one or two sentences.]** Future work comprises the modeling of the hydraulic actuators, gain scheduling over the stroke, the inclusion of lift-off loads and Hardware-in-the-Loop tests leading to a scaled prototype at sea.

Acknowledgements

The authors thank TECSpace and the School of Electronics Engineering of the Instituto Tecnológico de Costa Rica for supporting the MARLUP project, and the members of earlier MARLUP stages whose work is referenced here. **[TODO: complete acknowledgements and funding.]**

References

(a) Software-in-the-Loop



(b) Processor-in-the-Loop

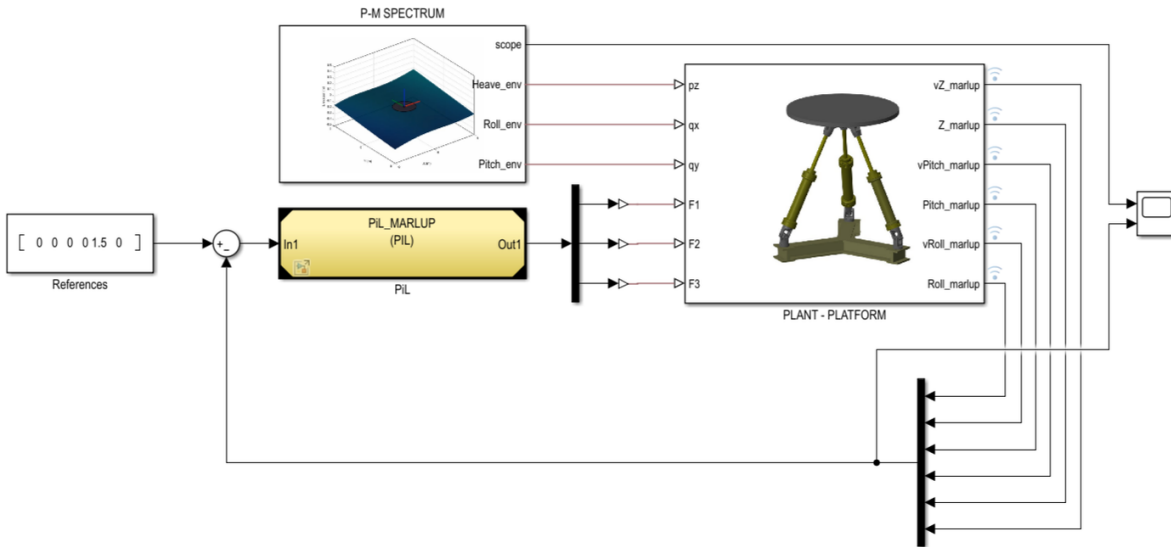


Fig. 11: Simulink harness in the two verification configurations. (a) SiL: the PiL_MARLUP subsystem runs as C code compiled for the host. (b) PiL: the same code cross-compiled and executed on the microcontroller. Everything outside that block, namely the plant, the P-M wave generator, the sensor models and the reference, is identical in both.

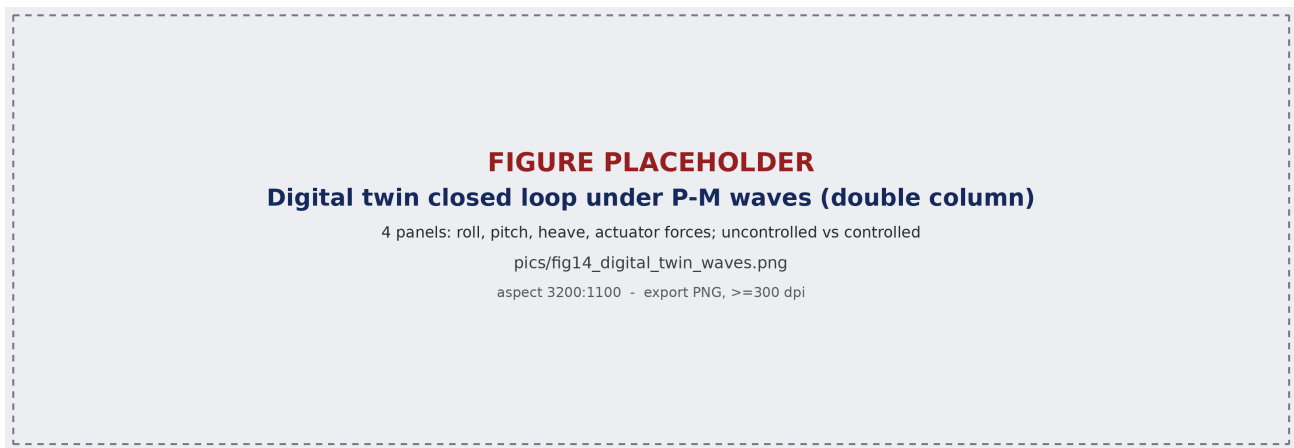


Fig. 15: Closed-loop digital twin under P–M waves: roll, pitch, heave and actuator forces, with the uncontrolled base motion shown for reference.